

# Hardware Qualification ISO - Setup and usage instructions

Fedora hardware validation at Meta consists of a few automated and manual tests performed on the hardware by the Client Platform Engineering team (CPE). These tests are a combination of known open-source stress tests and benchmarks as well as certain manual tests performed to confirm certain components work as expected.

In order for hardware validation testing to be simplified and performed on a Fedora Vanilla installation that, besides internal Meta management tooling, matches how devices are generally used, we created this ISO (fedora-fantasy-hwqual.iso).

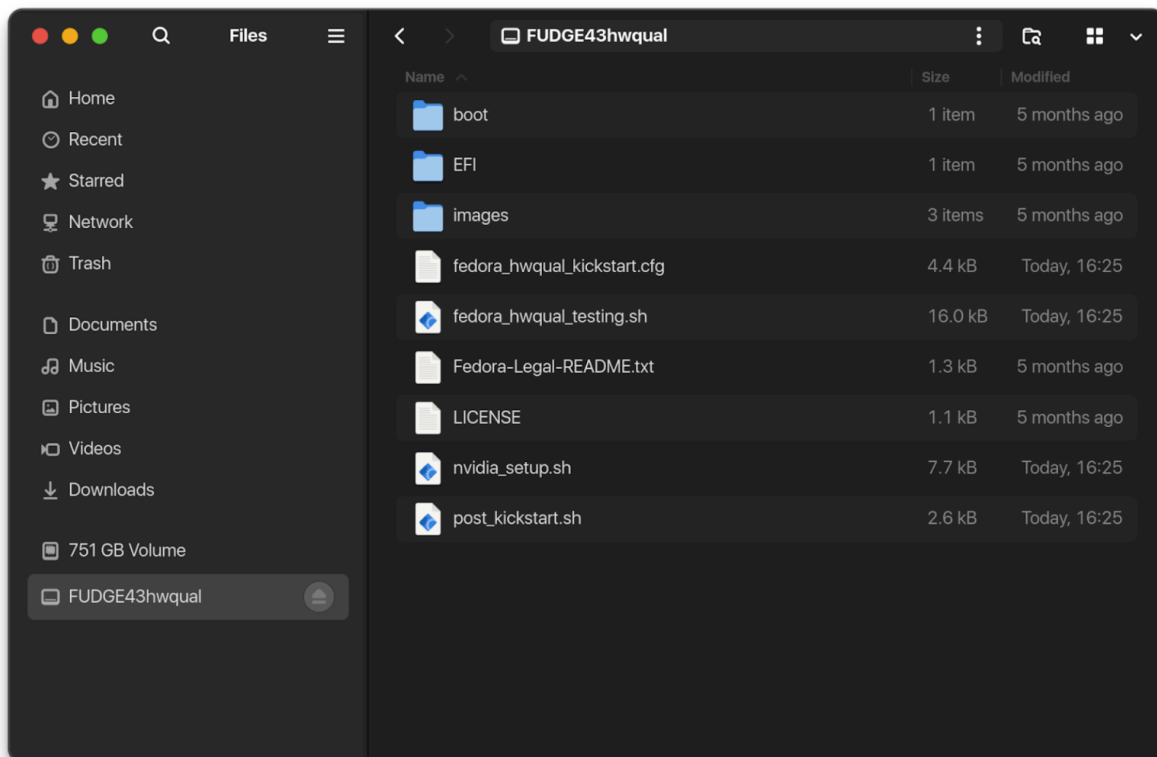
## Automated testing with custom ISO

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### ISO structure

The ISO is generated through automation and built on top of the Fedora Everything netinstaller ISO as found on public Fedora repositories here: <https://fedoraproject.org/misc/#everything>

We include in the ISO the following 4 files which can also be found on the root path of the ISO when mounted.



## fedora\_hwqual\_kickstart.cfg

Kickstart is an answer file which can be provided to a netinstaller and automate the steps otherwise performed by user selections (for example, system language, desktop environment etc)

This kickstart roughly does the following:

- Defines the primary source repo to get official Fedora packages
- Answers some automation basics like language, region etc.
- Partitions a drive with a 2GB EFI partition, a 2GB EXT4 partition for `/root` and the rest of the drive as BTRFS
- Encrypts the drive with the simple passphrase “fedora”
- Installs the packages needed for `@^workstation-product-environment`
  - (this is the default workstation option when using the Fedora live Gnome ISO too)
- If all goes well, preps `post_kickstart` to start next.

Kickstart then goes to `fedora_stable_post_kickstart.sh`.

## post\_kickstart.sh

This is a script that is executed in the `%post` section of the anaconda kickstart flow which allows us to run certain steps additionally to a typical kickstart install.

Post kickstart does the following:

- Moves the scripts baked into the ISO to the `/usr/local/bin/` system path
- Updates the kernel to the latest version as well as all system updates.
- Runs some additional package installs with `dnf`, like Google Chrome, benchmark tooling like `phoronix-test-suite`, `glmark2` etc.
- Creates a randomly generated file used as a secondary LUKS decryption key so on reboot, the password prompt can be skipped
- Executes the next script bundled in the ISO to install Nvidia drivers if needed

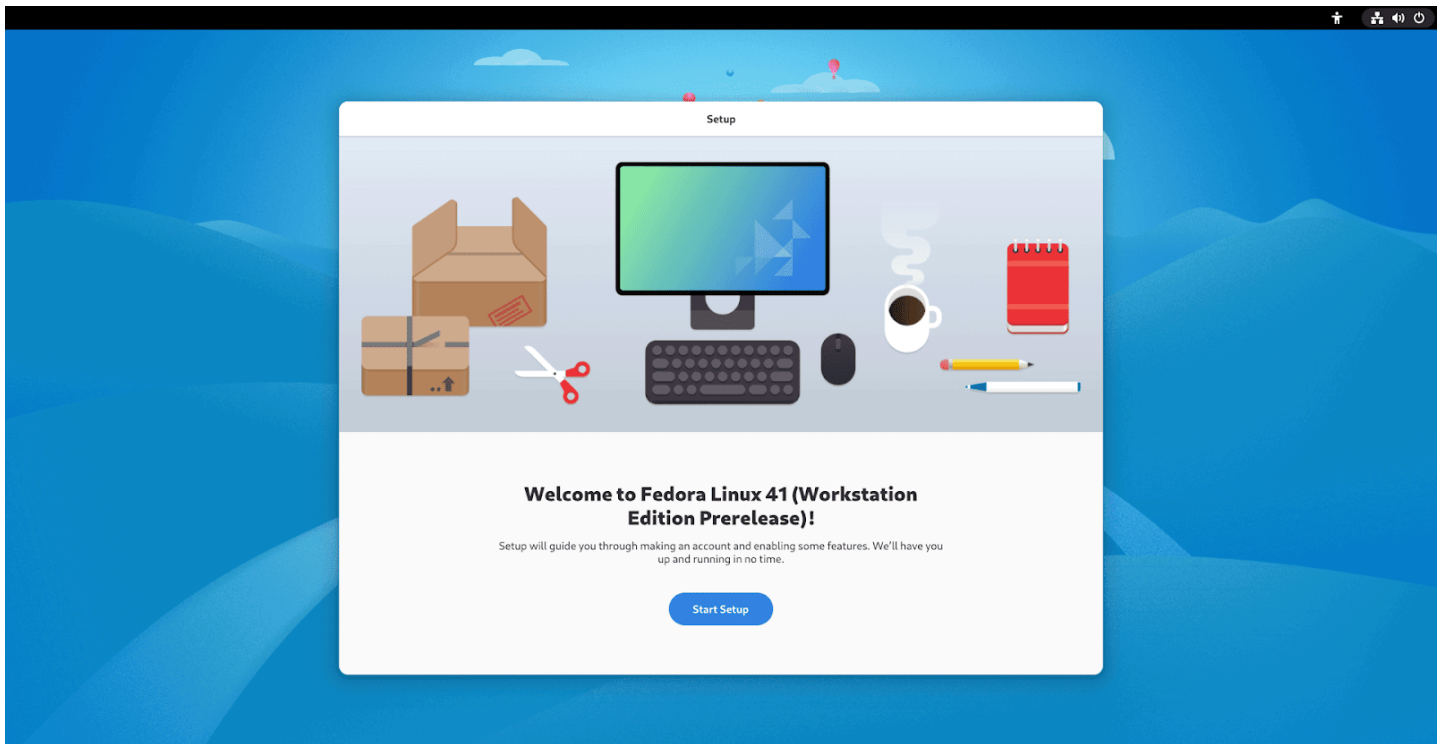
## nvidia\_setup.sh

This script will handle Nvidia driver installation if Nvidia hardware is detected on the system. It can run under two different modes, `rpmfusion` and `nvidia`.

The first mode will install Nvidia drivers by enabling 3rd party rpmfusion repositories and use the driver provided by that repo and the open source kernel modules handled by `akmod`. This option is also the default during automated installer since it's the preferred driver Meta and CPE use and test with,

The second mode will install the official Nvidia driver from Nvidia's cuda repository, using dkms to manager the provided kernel module. This is an option we provide to our Linux users if they require cuda libraries for development and they can uninstall the rpmfusion drivers and re-execute this script manually to switch to the official drivers instead of the rpmfusion ones.

Once all the above finish executing, the device will automatically restart and the Welcome to Fedora OOB screen will appear:



## fedora\_hwqual\_testing.sh

This script is also added to the ISO but not executed automatically. It will be moved by the kickstart phase to `/usr/local/bin/` so it can be called from the terminal

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## Benchmark script design

The script is a collection of benchmark and stress tests separated into 5 main categories.

CPU, GPU, RAM, Drive, OS

Under each category, a collection of tests will be performed as follows:

### CPU

1. 5 Minute CPU stress test with `stress` on all cores
2. `cachebench` through the phoronix-test-suite
3. `Coremark` through the phoronix-test-suite
4. `Graphics-magick` through the phoronix-test-suite
5. `Epoch` through the phoronix-test-suite
6. `Deepspeech` through the phoronix-test-suite

## GPU

1. `furmark` stress test through phoronix-test-suite
2. `glmark2-wayland` benchmark
3. Unigine Superposition through phoronix-test-suite
4. `blender` off-screen rendering through phoronix-test-suite (also includes cpu testing)

Note here that before running any of these tests, the script will attempt to detect if the system uses Dual / Hybrid GPUs and Nvidia like certain laptops have and if so, attempt to run the tests against the discrete GPU rather than the primary running the gnome session.

## RAM

1. `memtester` stress test against 80% of the free system RAM
2. `sysbench memory` speed test
3. `tinymembench` through the phoronix-test-suite
4. `ramspeed` through the photonix-test-suite

## Drive

1. `sysbench fileio` benchmark against 15G size page files
2. `iozone` benchmark
3. `hdparm` speedtest
4. `fio` speedtest through the phoronix-test-suite
5. `dbench` speedtest through the phoronix-test-suite

## Operating System

- Basic operating system functions testing with `osbench` through the phoronix-test-suite

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## Final report

Once all tests are performed an HTML file report will be generated and saved in the user Documents folder, which will create a summary of the device specifications and the tests performed. The Device specification will be gathered with the tool `inxi`. The final report will look like this:

Hardware Stress Testing and Benchmarks Report

Start Time: Wed 25 Mar 2026 6:26:39

| System Info |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Test Results      |                                                             |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Category    | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Category          | Test                                                        | Sample Result                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                  |
| System      | System:<br>Host: angelotheo-fedora-PW0MBZL0 Kernel: 6.18.16-200.fc43.x86_64<br>arch: x86_64 bits: 64<br>Console: pts/pts/3 Distro: Fedora Linux 43 (Workstation Edition)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPU Stress Test   | stress                                                      | stress: info: [677046] successful run completed in 300s                                                                                                                                                                                                                                                                                                                                                |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | cachebench                                                  | Average: 82775.417784 MB/s                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                  |
| Machine     | Machine:<br>Type: Laptop System: LENOVO product: 21TES0BQ00 v: ThinkPad T1g Gen 8<br>serial: PW0MBZL0<br>Mobo: LENOVO model: 21TES0BQ00 v: SDK0T76530 WIN serial: W1HV58E10BL<br>Firmware: UEFI vendor: LENOVO v: N4EET21W (1.07 ) date: 12/29/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CPU Benchmark     | coremark                                                    | Average: 336381.793335 Iterations/Sec                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | graphics-magick                                             | Average: 64 Iterations Per Minute                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                  |
| CPU         | CPU:<br>Info: 16-core model: Intel Core Ultra 7 265H bits: 64 type: MCP cache:<br>L2: 28 MiB<br>Speed (MHz): avg: 1583 min/max: 400/5300;4600;2500 cores: 1: 1583 2: 1583<br>3: 1583 4: 1583 5: 1583 6: 1583 7: 1583 8: 1583 9: 1583 10: 1583 11: 1583<br>12: 1583 13: 1583 14: 1583 15: 1583 16: 1583                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   | epoch                                                       | Epoch3D Deck: Cone:<br>688.53<br>Average: 688.53 Seconds                                                                                                                                                                                                                                                                                                                                               |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | deepspeech        | Acceleration: CPU:<br>100.8183<br>Average: 100.8183 Seconds |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                  |
| Graphics    | Graphics:<br>Device-1: Intel Arrow Lake-P [Arc Pro 130T/140T] driver: i915 v: kernel<br>Device-2: NVIDIA GB206M [GeForce RTX 5070 Max-Q / Mobile] driver: nvidia v: 580.126.18<br>Device-3: Luxvisions Innotech Integrated Camera<br>driver: hid-sensor-hub,usbhid,uvcvideo type: USB<br>Display: unspecified server: X.Org v: 24.1.9 with: Xwayland v: 24.1.9<br>driver: gpu: i915 resolution: 3840x2400-120Hz<br>API: OpenGL v: 4.6.0 vendor: nvidia v: 580.126.18 renderer: NVIDIA GeForce RTX 5070 Laptop GPU/PCIe/SSE2<br>API: EGL Message: EGL data requires eglinfo. Check --recommends.<br>Info: Tools: api: glxinfo gpu: nvidia-settings,nvidia-smi x11: xdrinfo, xdpinfo, xprop, xrandr                                                                                                                                                                                                                                                                                                                                   | GPU Stress Test   | furmark                                                     | Average: 28.097122 FPS                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GPU Benchmark     | glmark2-wayland                                             | glmark2 Score: 2321                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | unigine-super                                               | Average: 44.2 Frames Per Second                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                  |
| Memory      | Memory:<br>System RAM: total: 64 GiB available: 62.21 GiB used: 8.01 GiB (12.9%)<br>Array-1: capacity: N/A slots: 8 note: check modules: 8 EC: None<br>Device-1: Controller0-ChannelA-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-2: Controller0-ChannelB-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-3: Controller0-ChannelC-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-4: Controller0-ChannelD-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-5: Controller1-ChannelA-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-6: Controller1-ChannelB-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-7: Controller1-ChannelC-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s<br>Device-8: Controller1-ChannelD-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s | GPU Benchmark     | blender                                                     | Blend File: BMW27 - Compute: CPU-Only:<br>111.07<br>Blend File: Junkshop - Compute: CPU-Only:<br>149.02<br>Blend File: Classroom - Compute: CPU-Only:<br>285.85<br>Blend File: Fishy Cat - Compute: CPU-Only:<br>163.82<br>Blend File: Barbershop - Compute: CPU-Only:<br>1172.33<br>Blend File: Pabellon Barcelona - Compute: CPU-Only:<br>379.16                                                     |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | RAM Stress Test                                             | memtester                                                                                                                                                                                                                                                                                                                                                                                              | Done.                                                                                            |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |                                                             | sysbench memory                                                                                                                                                                                                                                                                                                                                                                                        | Total operations: 50081278 (5007477.33 per second)<br>56720.00 MiB transferred (5671.36 MiB/sec) |
| Audio       | Audio:<br>Device-1: Intel Arrow Lake cAVS driver: sof-audio-pci-intel-mtl<br>API: ALSA v: k6.18.16-200.fc43.x86_64 status: kernel-api                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | RAM Benchmark     | tinymembench                                                | Average: 58410.8 MB/s<br>Average: 58410.8 MB/s                                                                                                                                                                                                                                                                                                                                                         |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | ramspeed                                                    | Average: 32133.87 MB/s                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                  |
| Network     | Network:<br>Device-1: Intel Arrow Lake CNWi WiFi driver: iwlwifi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Drive Stress Test | sysbench fioio                                              | reads/s: 399.65<br>writes/s: 266.44<br>fsyncs/s: 852.81<br>read, MiB/s: 6.24<br>written, MiB/s: 4.16<br>total time: 300.1083s                                                                                                                                                                                                                                                                          |                                                                                                  |
| Bluetooth   | Bluetooth:<br>Device-1: Intel driver: btusb type: USB<br>Report: btmgmt ID: hci0 state: up address: C8:95:CE:A5:8D:7B bt-v: 5.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | iozone                                                      | Avg throughput per process = 1191302.50 kB/sec                                                                                                                                                                                                                                                                                                                                                         |                                                                                                  |
| Drives      | Drives:<br>Local Storage: total: 953.87 GiB used: 100.01 GiB (10.5%)<br>ID-1: /dev/nvme0n1 vendor: Samsung model: MZVLCT10HFLU-00BLL<br>size: 953.87 GiB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   | hdparm                                                      | Timing cached reads: 26876 MB in 1.99 seconds = 13487.78 MB/sec<br>Timing buffered disk reads: 12694 MB in 3.00 seconds = 4230.87 MB/sec                                                                                                                                                                                                                                                               |                                                                                                  |
| Partitions  | Partition:<br>ID-1: / size: 249.98 GiB used: 99.18 GiB (39.7%) fs: btrfs dev: /dev/dm-0<br>ID-2: /boot size: 1.9 GiB used: 816.6 MiB (42.0%) fs: ext4 dev: /dev/nvme0n1p2<br>ID-3: /boot/efi size: 2 GiB used: 38.9 MiB (1.9%) fs: vfat dev: /dev/nvme0n1p1<br>ID-4: /home size: 249.98 GiB used: 99.18 GiB (39.7%) fs: btrfs dev: /dev/dm-0<br>ID-5: /var/log size: 249.98 GiB used: 99.18 GiB (39.7%) fs: btrfs dev: /dev/dm-0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Drive Benchmark   | fio                                                         | 75.2<br>19300<br>1182<br>303000<br>Average: 646 MB/s                                                                                                                                                                                                                                                                                                                                                   |                                                                                                  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   | dbench                                                      | Average: 2310.04 MB/s                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |
| Battery     | Battery:<br>ID-1: BAT0 charge: 87 Wh (96%) condition: 90.7/90 Wh (100.8%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | System Benchmark  | osbench                                                     | Test: Read:<br>Average: 8749.905068 MB/s<br>Test: Write:<br>Average: 62057.059385 MB/s<br>Test: Create Files:<br>Average: 18.803995 us Per Event<br>Test: Create Threads:<br>Average: 11.229515 us Per Event<br>Test: Launch Programs:<br>Average: 41.909218 us Per Event<br>Test: Create Processes:<br>Average: 35.67934 us Per Event<br>Test: Memory Allocations:<br>Average: 73.136091 Ns Per Event |                                                                                                  |

Completion time: Wed 25 Mar 2026 6:26:42

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full Log | Updating and loading repositories:<br>Allowed part of fb-runtime prod repo 100%   764.0 B/s   421.0 B   00m01s<br>Facebook packages for any Fedora relea 100%   760.0 B/s   419.0 B   00m01s<br>Repositories loaded.<br>Package "phoronix-test-suite-10.8.4-10.fc43.noarch" is already installed.<br>Package "glmark2-2023.01*20250221gitcebb63-2.fc43.x86_64" is already installed.<br>Package "sysbench-1.0.20-19.fc43.x86_64" is already installed.<br>Package "memtester-4.7.1-2.fc43.x86_64" is already installed.<br>Package "iozone-3.506-5.fc43.x86_64" is already installed.<br>Package "hdparm-9.65-9.fc43.x86_64" is already installed.<br>Package "stress-1.0.7-7.fc43.x86_64" is already installed. |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

The complete log of all tests performed will be included in the HTML report but is also written here:  
[/root/hardware\\_validation.log](#)

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## Manual testing

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Additional testing performed besides the automated tests with the ISO above are documented below. These tests are usually performed by a CPE engineer on a provisioned device as internally managed which will include our continuous management and security tooling.

## Tools needed

- GPU Tests
  - [glmark2](#)
  - [Blender](#)
  - [Unigine](#)
- CPU tests
  - [GTKStressTesting](#)
  - [Geekbench](#)
- RAM test
  - [memtester](#) (`sudo dnf install memtester`)
  - [sysbench](#) (`sudo dnf install sysbench`)
- battery test
  - [GNOME power statistics](#) (`sudo dnf install gnome-power-manager`)
- Drive test
  - [KDiskMark](#) (`flatpak install flathub io.github.jonmagon.kdiskmark`)
- Device specs tool
  - [inxi](#) (`sudo dnf install inxi`)

## Additional tooling to consider

- Hardware probe (hw-probe): [https://fedoraproject.org/wiki/Hardware\\_probe](https://fedoraproject.org/wiki/Hardware_probe)
    - Example output: <https://linux-hardware.org/?probe=273526bab2>
- 

## Tests to perform

### Hardware tests:

- GPU tests
  - [glmark2](#)
    - `glmark2-wayland --fullscreen`

- Capture score output and compare with others online
  - [Blender](#)
    - Ensure output matches overall GPU score from others online
  - [Unigine](#)
    - Ensure test runs at ultra graphics settings with expected score
    - No visible screen tearing
- CPU tests:
  - [GTKStressTesting](#) (`flatpak install flathub com.leinardi.gst`)
    - System can maintain stability under 100% load for 30 mins
  - [Geekbench](#)
    - Performance scores match the hardware expectations
- Memory:
  - [memtester](#)
    - Example command: `sudo memtester 16G 3`
  - [sysbench](#)
    - Example command: `sysbench memory run`
- Disk: [KDiskMark](#) (`flatpak install flathub io.github.jonmagon.kdiskmark`)
  - SSD benchmark matches hardware expectations
    - Ensure CoW is disabled
- battery: GNOME power statistics
  - For expected power consumption do the following:
    - Set power mode to Power Saver
    - Screen brightness to 50%
    - Caffeine to keep device awake
    - Run CPU stress test above to keep CPU usage at 15-20% (2-4 workers should suffice)
- External ports - USB ports - charging
- Webcam: Camera app should be enough
- Microphone: <https://zoom.us/test>
- Audio Input: System settings Microphone input gauge or zoom test call.
- Audio Output: System setting audio out test or zoom test call.
- WiFi: system settings - Wi-Fi - connect to networks
- Bluetooth: system settings - Bluetooth - Connect to a bluetooth device
- Keyboard
  - During preboot (BIOS), LUKS and OS
- Mouse/Trackpad
  - During preboot (BIOS), LUKS and OS

#### Software tests:

- Vanilla OS installation
  - Any issues from the out of the box experience
  - Driver support
  - DNF update - LVFS firmware support (`fwupdmgm update`)
- Suspend/sleep - resume
- System settings functionality
  - Function keys
    - Change screen brightness
    - Volume control
  - Turn WiFi/Bluetooth on/off

### Internal tools:

- provisioning
- continuous device management
- certificate handling
- yum repo / package distribution

### Final go/no-go test:

- Spend a week using the device as the daily driver
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## Results

We publish the results in our internal Wiki for reference by all our employees. Results are captured in a spreadsheet with a specific template.

Additionally, we create and link the following 2 reports to the wiki so we have the exact specs of the devices qualified:

- output of `sudo inxi -e`
- output of `sudo -E hw-probe -all -upload` like here:  
<https://linux-hardware.org/?probe=a98b6568d5>